# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 20 | Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. |
| 2 | **(a)** -14 | Put $p(z) = 0$. |
| 3 | **(b)** $\frac{39}{65}$ | Reduce the fraction; only 2 and 5 may remain in the denominator. |
| 4 | **(a)** $49x^2 + 56xy + 16y^2$ | Use the square identity. |
| 5 | **(b)** $(4.5,\ 4.5)$ | Average the coordinates. |
| 6 | **(b)** 3 | Slope $= \frac{y_2 - y_1}{x_2 - x_1}$. |
| 7 | **(d)** 4 | $400 = 2^4 \times 5^2$ |
| 8 | **(a)** $6x^2 + 19x + 15$ | Multiply term by term. |
| 9 | **(b)** $(-1, -3)$ | The centre is the midpoint of a diameter. |
| 10 | **(c)** -3 | Put $y = 0$. |
| 11 | **(b)** $\frac{4}{33}$ | $x = \frac{ab}{99}$, then reduce. |
| 12 | **(b)** 45 | $a^2 + b^2 = (a - b)^2 + 2ab$. |
| 13 | **(a)** $(3, 0)$ | Set $PA = PB$ with $P = (x, 0)$. |
| 14 | **(b)** 4 | Substitute $y$ and solve for $x$. |
| 15 | **(a)** 6 | Find how many digits repeat. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $d^2 = (4 - (1))^2 + (5 - (1))^2 = 25$ (1); $d = 5$ units (1). | 1+1 |
| 17 | $m = \frac{13 - (4)}{5 - (2)}$ (1); $m = 3$ (1). | 1+1 |
| 18 | Multiply by 10 and by 100 and subtract: $90x = 75$ (1); $x = \frac{5}{6}$ (1). | 1+1 |
| 19 | Multiply term by term (1); $= 6x^2 + 17x + 5$ (1). | 1+1 |
| 20 | $\left(\frac{3+(6)}{2}, \frac{-4+(7)}{2}\right)$ (1); Midpoint $= (4.5,\ 1.5)$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | $AB = 1 \times 5 = 5$ (direction $(3, 4)$); $BC = 2 \times 5 = 10$ (1); $AC = 15 = AB + BC$ (1); So the points are collinear (1). | 1+1+1 |
| 22 | $18 = m(2) + 12$ (1); $m = 3$ (1); $x$-intercept: $0 = 3x + 12$, $x = -4$ (1). | 1+1+1 |
| 23 | $1000x - x = 27$ (1); $999x = 27$ (1); $x = \frac{1}{37}$ (1). | 1+1+1 |
| 24 | $x^2 + \frac{1}{x^2} = (6)^2 - 2 = 34$ (1); $\left(x - \frac{1}{x}\right)^2 = 34 - 2 = 32$ (2). | 1+2 |
| | **OR:** $(2 - k)^2 + 16 = 25$ (1); $(2 - k)^2 = 9$ (1); $k = -1$ or $k = 5$ (1). | 1+1+1 |
| 25 | Points $(0, 5)$, $(1, 1)$, $(2, -3)$ (1); Slope $= \frac{-3 - (5)}{2 - 0}$ (1); $= -4$ (1). | 1+1+1 |
| | **OR:** $x = 1 + 0.2\overline{3}$ (1); $0.2\overline{3} = \frac{7}{30}$ (1); $x = \frac{37}{30}$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | $AB^2 = 9 + 16 = 25$ (1); $BC^2 = DA^2 = CD^2 = 25$: all sides equal 5 (1); $AC = 2 \times 3 = 6$ (1); $BD = 2 \times 4 = 8$ (1); $6 \neq 8$, so not a square (1). | 1+1+1+1+1 |
| | **OR:** Slope $= \frac{16 - (8)}{3 - (1)} = 4$ (1); $p(x) = 4x + 4$ (1); Zero: $x = -1$ (1); $p(5) = 24$ (2). | 1+1+1+2 |
| 27 | Let $x = 1.1\overline{6}$; $10x$ and $100x$ give $90x = 105$ (1); $x = \frac{7}{6}$ (1); Long division: $1.1\overline{6}$ (1); Matches the given decimal (2). | 1+1+1+2 |
| | **OR:** $x^2 - y^2 = (x + y)(x - y)$ (1); $= 12 \times 4 = 48$ (1); $4xy = (x + y)^2 - (x - y)^2 = 144 - 16$, so $xy = 32$ (1); $2(x^2 + y^2) = (x + y)^2 + (x - y)^2 = 160$ (1); $x^2 + y^2 = 80$ (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 7 | Square both sides. |
| 2 | **(b)** Quadrant III | Use the sign conditions on the coordinates. |
| 3 | **(b)** $8x - 5$ | Add like terms. |
| 4 | **(d)** 6 | Multiply under the root. |
| 5 | **(c)** 1006009 | Write $1003 = 1000 + 3$ |
| 6 | **(c)** 36 | Add the three side lengths. |
| 7 | **(b)** 4 | $p(2) - p(1) = a$. |
| 8 | **(b)** rational | Collect the rational and irrational parts. |
| 9 | **(c)** 5600 | Use $a^2 - b^2 = (a+b)(a-b)$. |
| 10 | **(a)** $(-4, -9)$ | Both signs change. |
| 11 | **(a)** 82 | Substitute $n$. |
| 12 | **(d)** $\frac{3}{7}$ | Take the average. |
| 13 | **(d)** $(x-7)(x+4)$ | Find the pair of numbers. |
| 14 | **(c)** 15 | Area = length × breadth. |
| 15 | **(d)** parallel | Use the slope-intercept form. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $p - q = 3x + 9$ (1); Zero $x = -3$ (1). | 1+1 |
| 17 | Long division of 5 by 22 (1); $= 0.2\overline{27}$, block 27 (1). | 1+1 |
| 18 | $x^2 + y^2 = (x+y)^2 - 2xy$ (1); $= 121 - 56 = 65$ (1). | 1+1 |
| 19 | Centre is the midpoint of the diameter (1); $B = (-3, -6)$ (1). | 1+1 |
| 20 | $a_1 = 5$, $a_2 = 13$, $a_3 = 21$ (1); $a_{10} = 77$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | $a^2 + b^2 = 4^2 + 2(21)$ (1); $= 58$ (1); $(a+b)^2 = 16 + 4(21) = 100$ (1). | 1+1+1 |
| 22 | $AB^2 = 18$, $BC^2 = 18$; $CA^2 = 36$ (1); Two sides equal (1); $AB^2 + BC^2 = CA^2$: right angle at $B$ (1). | 1+1+1 |
| 23 | $a_n = 7n - 1$ (1); $7n - 1 = 90$ (1); $n = 13$ (1). | 1+1+1 |
| 24 | Lowest terms: $\frac{7}{25}$ (1); Denominator $= 25 = 5^2$ (1); Terminates, since only 2 and 5 occur (1). | 1+1+1 |
| | **OR:** Split the middle term: $11x = 8x + 3x$ (1); Group the terms (1); $= (2x+3)(x+4)$ (1). | 1+1+1 |
| 25 | $AB = 2 \times 13 = 26$ (direction $(5, 12)$); $BC = 1 \times 13 = 13$ (1); $AC = 39 = AB + BC$ (1); So the points are collinear (1). | 1+1+1 |
| | **OR:** $32 = m(3) + 20$ (1); $m = 4$ (1); $x$-intercept: $0 = 4x + 20$, $x = -5$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | $D = A + C - B = (2, 5)$ (1); Midpoint of $AC = (3.5, 3)$ (1); Midpoint of $BD = (3.5, 3)$ (same point) (1); Diagonals bisect each other (1); Hence $ABCD$ is a parallelogram (1). | 1+1+1+1+1 |
| | **OR:** $E(n) = 150n + 5000$ (1); $E(20) = 8000$ (1); $150n + 5000 = 14000$, $n = 60$ (1); Extra items $= \frac{1200}{150} = 8$ (2). | 1+1+1+2 |
| 27 | $(3 + \sqrt{5})^2 = 14 + 6\sqrt{5}$ (1); $(3 - \sqrt{5})^2 = 14 - 6\sqrt{5}$ (1); Sum $= 28$ (1); The surd terms cancel (1); Result is rational (1). | 1+1+1+1+1 |

| | | |
|---|---|---|
| | **OR:** $(a+b)^2 = a^2 + b^2 + 2ab$ (1); $81 = a^2 + b^2 + 28$ (1); $a^2 + b^2 = 53$ (1); $(a-b)^2 = (a+b)^2 - 4ab = 81 - 56 = 25$ (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 11 | Apply Brahmagupta's rules in order. |
| 2 | **(a)** 32 | $4xy = (x+y)^2 - (x-y)^2$. |
| 3 | **(c)** 65 | Use $d^2 = (x_2-x_1)^2 + (y_2-y_1)^2$. |
| 4 | **(a)** -30 | Put $p(z) = 0$. |
| 5 | **(c)** $\frac{21}{70}$ | Reduce the fraction; only 2 and 5 may remain in the denominator. |
| 6 | **(d)** $36x^2 - 60xy + 25y^2$ | Use the square identity. |
| 7 | **(c)** $(-1.5,\ 6.5)$ | Average the coordinates. |
| 8 | **(c)** 3 | Slope $= \frac{y_2-y_1}{x_2-x_1}$. |
| 9 | **(a)** 3 | $250 = 2^1 \times 5^3$ |
| 10 | **(d)** $6x^2 + 25x + 14$ | Multiply term by term. |
| 11 | **(a)** $(3, 0)$ | The centre is the midpoint of a diameter. |
| 12 | **(b)** 2 | Put $y = 0$. |
| 13 | **(c)** $\frac{6}{11}$ | $x = \frac{ab}{99}$, then reduce. |
| 14 | **(c)** 73 | $a^2 + b^2 = (a-b)^2 + 2ab$. |
| 15 | **(d)** $(-2, 0)$ | Set $PA = PB$ with $P = (x, 0)$. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $= 8^2 - (\sqrt{5})^2$ (1); $= 59$ (1). | 1+1 |
| 17 | $1002 = 1000 + 2$ (1); $1002^2 = 1004004$ (1). | 1+1 |
| 18 | $d^2 = (2-(-2))^2 + (6-(3))^2 = 25$ (1); $d = 5$ units (1). | 1+1 |
| 19 | $m = \frac{8-(0)}{3-(-1)}$ (1); $m = 2$ (1). | 1+1 |
| 20 | Multiply by 10 and by 100 and subtract: $90x = 55$ (1); $x = \frac{11}{18}$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | $1000x - x = 135$ (1); $999x = 135$ (1); $x = \frac{5}{37}$ (1). | 1+1+1 |
| 22 | $x^2 + \frac{1}{x^2} = (8)^2 - 2 = 62$ (1); $\left(x - \frac{1}{x}\right)^2 = 62 - 2 = 60$ (2). | 1+2 |
| 23 | $(-3-k)^2 + 144 = 169$ (1); $(-3-k)^2 = 25$ (1); $k = -8$ or $k = 2$ (1). | 1+1+1 |
| 24 | Points $(0,-6)$, $(1,3)$, $(2,12)$ (1); Slope $= \frac{12-(-6)}{2-0}$ (1); $= 9$ (1). | 1+1+1 |
| | **OR:** $x = 2 + 0.4\overline{6}$ (1); $0.4\overline{6} = \frac{7}{15}$ (1); $x = \frac{37}{15}$ (1). | 1+1+1 |
| 25 | $a^2 + b^2 = 6^2 + 2(40)$ (1); $= 116$ (1); $(a+b)^2 = 36 + 4(40) = 196$ (1). | 1+1+1 |
| | **OR:** $AB^2 = 18$, $BC^2 = 18$; $CA^2 = 36$ (1); Two sides equal (1); $AB^2 + BC^2 = CA^2$: right angle at $B$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | $AB^2 = 25 + 144 = 169$ (1); $BC^2 = DA^2 = CD^2 = 169$: all sides equal 13 (1); $AC = 2 \times 5 = 10$ (1); $BD = 2 \times 12 = 24$ (1); $10 \neq 24$, so not a square (1). | 1+1+1+1+1 |
| | **OR:** Slope $= \frac{20-(5)}{6-(3)} = 5$ (1); $p(x) = 5x - 10$ (1); Zero: $x = 2$ (1); $p(4) = 10$ (2). | 1+1+1+2 |
| 27 | Let $x = 2.3\overline{3}$; $10x$ and $100x$ give $90x = 210$ (1); $x = \frac{7}{3}$ (1); Long division: $2.\overline{3}$ (1); Matches the given decimal (2). | 1+1+1+2 |
| | **OR:** $x^2 - y^2 = (x+y)(x-y)$ (1); $= 10 \times 6 = 60$ (1); $4xy = (x+y)^2 - (x-y)^2 = 100 - 36$, so $xy = 16$ (1); $2(x^2 + y^2) = (x+y)^2 + (x-y)^2 = 136$ (1); $x^2 + y^2 = 68$ (1). | 1+1+1+1+1 |

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